

Fast Analytics for NYC Taxi Data

TEAM:
JUTTU GAJENDRA ANURAG





Use Cases

COMMUTER

Commuters can query the fastest routes to save time, receiving optimized trip suggestions based on real-time traffic data analytics.

TRANSPORTATION ANALYST

Transportation analysts analyze traffic patterns to identify congestion hotspots, using insights to recommend improvements for urban transit efficiency.

CITY PLANNER

City planners review trip density by zones, allowing them to plan infrastructure upgrades that enhance urban mobility and accessibility.



Methodology Overview

DATA TRANSFORMATION

The collected data is cleaned and transformed to prepare it for analytical queries, enhancing its accuracy and usability for stakeholders.

QUERY OPTIMIZATION

We implement and optimize queries within the Scala and Spark framework to ensure performance and efficiency during data retrieval operations.

Data Sources and Fields

PRIMARY DATASET

The primary source for our analysis is the **NYC Taxi Trip Records**, consisting of over **1,000,000** rows of trip data.

KEY FIELDS

The dataset includes essential fields such as **pickup/dropoff timestamps**, **location IDs**, **trip distance**, **fare amount**, and **passenger count**.

Milestones Overview

Week 1

In the first week, we will set up the development environment and thoroughly understand the data schema to ensure a solid foundation for future work.

Week 2

During the second week, we will focus on data ingestion, cleaning, and preprocessing to prepare the NYC Taxi Trip data for effective analytics and accurate results.

Week 3

The third week will involve implementing various analytics and query strategies, transforming the cleaned data into meaningful insights regarding taxi trip patterns and metrics.

Week 4

In the final week, we will conduct testing and validation of our queries, ensuring performance benchmarks are met, and prepare our findings for the final presentation.

Technical Implementation Overview

CORE COMPONENTS

Scala serves as the primary programming language, facilitating data ingestion, preprocessing, and execution of analytical queries efficiently throughout the project.

OPTIONAL TOOLS

SQL may be utilized for complex query executions, while Python can assist in visualizing data results, enhancing the overall analysis experience.



Acceptance Criteria

DATA PROCESSED

We aim to process **at least 1,000,000** rows of taxi trip data to ensure robust analytics and meaningful insights.

QUERY TYPES

The system will support a **minimum of 5 distinct query types**, enhancing flexibility in data analysis across various use cases.

QUERY SPEED

To achieve high performance, **95% of queries** must execute in under 5 seconds, ensuring quick access to vital information.



Project Goals

SCALABLE PIPELINE

Build a **robust analytics pipeline** using Scala to handle large datasets efficiently and effectively for real-time analytics.

MEANINGFUL QUERIES

Support impactful transportation-related queries that facilitate **data-driven decisions** in urban planning and improve commuter experiences.

HANDS-ON EXPERIENCE

Gain practical skills with **Spark and large datasets**, enhancing our technical expertise and ensuring readiness for future big data challenges.



Thank You for Your
Attention